

Derivation of Black Hole Universe Genesis

Based on the Spacetime Waterfall Model and Schwarzschild Geometry

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Abstract

This draft paper, inspired by an intuitive, hand-drawn conceptual map of spatial geometry, explores and derives the core mathematical framework of "Black Hole Cosmology." It utilizes the Gullstrand-Painlevé coordinates from General Relativity to explain the spacetime waterfall effect inside the event horizon; calculates the exact equivalence between the Schwarzschild radius and the Hubble radius of the observable universe; and finally, introduces spin torsion from the Einstein-Cartan theory to mathematically model the quantum bounce at the singularity, leading to subsequent infinite spatial expansion.

1 The Spacetime Waterfall Model and Superluminal Speeds Inside the Horizon

This section aims to prove that inside the event horizon of a black hole, space itself collapses toward the singularity at a speed greater than the speed of light, effectively trapping all photons (corresponding to the manuscript's notation: "light cannot swim, singularity sinking > speed of light").

In General Relativity, to describe the "flow" of space itself, we employ Gullstrand-Painlevé coordinates, commonly known as the "River model of black holes." Its line element is expressed as:

$$ds^2 = -c^2 dt_{gp}^2 + \left(dr + \sqrt{\frac{2GM}{r}} dt_{gp} \right)^2 + r^2 d\Omega^2 \quad (1)$$

From this equation, we can extract the radial component v_r , which represents the "flow velocity" of space:

$$v_r = -\sqrt{\frac{2GM}{r}} \quad (2)$$

where G is the gravitational constant, M is the mass of the black hole, and r is the radial distance from the singularity.

At the exact edge of the event horizon, the radius r equals the Schwarzschild radius $r_s = \frac{2GM}{c^2}$. Substituting this into the velocity equation yields:

$$|v_r| = \sqrt{\frac{2GM}{\frac{2GM}{c^2}}} = \sqrt{c^2} = c \quad (3)$$

This proves that at the boundary of the horizon, the speed of falling space is exactly equal to the speed of light c . When entering the interior of the black hole ($r < r_s$), the denominator becomes smaller, inevitably resulting in:

$$|v_r| > c \quad (4)$$

Conclusion: The mathematical proof supports the spacetime waterfall hypothesis. Inside the horizon, space falls toward the singularity faster than the speed of light. Photons (whose local "swimming" speed is always c) find themselves in a spacetime river flowing faster than c . Consequently, their net velocity remains directed toward the singularity, and any outward radiation is irreversibly swept inward.

2 Schwarzschild Cosmology: Equivalence of Universe Mass and Critical Radius

This section aims to prove that our observable universe is macroscopically and mathematically equivalent to a supermassive black hole (corresponding to the manuscript's question: "Are we living inside a black hole?"). We will calculate the "Schwarzschild radius" of our universe and compare it to its actual observable radius.

First, according to the Friedmann equations of the Big Bang model, assuming a flat universe, the critical density ρ_c is:

$$\rho_c = \frac{3H_0^2}{8\pi G} \quad (5)$$

where H_0 is the Hubble constant.

Second, the observational limit of our universe, known as the Hubble radius R_U , is defined as:

$$R_U = \frac{c}{H_0} \quad (6)$$

Assuming a uniform cosmic density, the total mass M_U of the observable universe is the product of its volume and density:

$$M_U = \frac{4}{3}\pi R_U^3 \rho_c \quad (7)$$

We then substitute the total mass of the universe M_U into the Schwarzschild radius formula for a black hole, $r_s = \frac{2GM}{c^2}$:

$$r_s = \frac{2G}{c^2} \left(\frac{4}{3}\pi \left(\frac{c}{H_0} \right)^3 \left(\frac{3H_0^2}{8\pi G} \right) \right) \quad (8)$$

Through algebraic simplification (the constants G , π , and 3 cancel out mutually):

$$r_s = \frac{2G}{c^2} \left(\frac{4\pi c^3}{3H_0^3} \cdot \frac{3H_0^2}{8\pi G} \right) = \frac{2G}{c^2} \left(\frac{c^3}{2G \cdot H_0} \right) = \frac{c}{H_0} = R_U \quad (9)$$

Conclusion: The mathematical calculation accurately demonstrates that the critical black hole radius r_s of our universe is exactly equal to the actual observable radius R_U of our universe. This result provides robust analytical evidence for the macroscopic hypothesis that "countless galaxies are enveloped within a giant black hole."

3 Quantum Bounce and Infinite Expansion at Extreme Density

This section explains why the singularity at the bottom of a black hole is not the physical end, but rather the trigger for a Big Bang-like event, creating an infinitely expanding internal space (corresponding to the manuscript's top notations: "Big Bang = Singularity" and "Infinite Enlargement").

In standard General Relativity, the density at the singularity becomes infinite, leading to mathematical divergence. To resolve this, we introduce the Einstein-Cartan Theory. This theory incorporates the "spin torsion" of fermions (the elementary particles that make up matter) into the geometric framework.

Under conditions of extreme high density, the spin torsion of fermions generates an incredibly powerful repulsive gravitational effect. The modified Friedmann Equation for cosmic expansion becomes:

$$H^2 = \frac{8\pi G}{3} \left(\rho - \frac{\rho^2}{\rho_{crit}} \right) \quad (10)$$

As matter continuously collapses, the density ρ increases and approaches the critical limit density ρ_{crit} (typically at the Planck density level). The negative term $-\frac{\rho^2}{\rho_{crit}}$ on the right side of the equation increases drastically, effectively neutralizing the standard gravitational term.

When $\rho = \rho_{crit}$, the expansion rate $H^2 = 0$, indicating that the collapse process halts instantaneously. Subsequently, the repulsive force dominates, causing matter to undergo a violent reverse expansion, known as a "Quantum Bounce."

Conclusion: The singularity is fundamentally a quantum bounce point triggered by an extreme-density repulsive mechanism. For an observer located inside this black hole, the moment of this bounce is experienced as a "Big Bang." The ensuing cosmic inflation flips the spacetime coordinates, and the internal spatial volume begins to expand exponentially over time, ultimately forming an inner universe capable of housing countless galaxies and expanding infinitely.